Learning Objective 2.1: I can describe the radiation behavior of simple resonant antennas (isotropic radiator, half-wave dipole, monopole, loop) and calculate their gain and impedance.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **A whip for the guard frequency.** You are asked to build a quarter-wave monopole for $f = 121.5$ MHz on a large aluminum ground plane, fed with $50\,\Omega$ coax. Take the thin-wire half-wave dipole as $73 + j42.5\,\Omega$.

   (a) How long is the whip?

$$\lambda = \frac{c}{f} = \frac{3\times 10^8}{121.5\times 10^6} = 2.47\text{ m}$$
$$L = \frac{\lambda}{4} = 0.617\text{ m}$$

L = $\boxed{0.617\text{ m}}$

   The whip is about 62 cm of metal above the plane.

   (b) State the input impedance you expect at the base of the whip, and explain in one or two sentences why it is not the dipole value.

   $Z_{\text{in}} = 36.5 + j21.3\,\Omega$, exactly half the dipole value. The image in the ground plane supplies the missing lower half of the dipole, so the current at the feed is the same as the dipole's, but the generator only drives half the structure and therefore only has to supply half the voltage. Half the voltage at the same current is half the impedance.

   (c) Find $|\Gamma|$ and the VSWR seen by the $50\,\Omega$ feed.

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} = \frac{-13.5 + j21.3}{86.5 + j21.3}$$
$$|\Gamma| = \frac{25.2}{89.1} = 0.283$$
$$\text{VSWR} = \frac{1 + 0.283}{1 - 0.283} = 1.79$$

VSWR = $\boxed{1.79}$

   Return loss $= -20\log_{10}(0.283) = 11.0$ dB .

   (d) You may change the whip or the ground plane, but you may not add parts. Give two changes that improve the match, and say what each one does to the impedance.

   (1) *Trim the whip* to about 0.24λ (roughly 59 cm). Shortening it a few percent tunes out the $+j21.3\,\Omega$ of reactance and leaves about $36\,\Omega$ real, so VSWR drops to $50/36 = 1.39$. The reactance, not the resistance, is what costs most of the mismatch. (2) *Droop the radials* of the ground plane down about $45°$. Tilting the radials raises the base resistance from roughly $36\,\Omega$ toward $50\,\Omega$, giving VSWR near 1.0 with no added components. This is why commercial ground-plane antennas have sagging radials.

2. **Which way do you lay the wire?** A perfectly conducting deck acts as an image plane. An antenna at height h above it radiates as the element pattern times a two-element array factor, $2|\cos(kh\cos\theta)|$ for a vertical element and $2|\sin(kh\cos\theta)|$ for a horizontal one, with θ measured from the vertical.

   (a) A horizontal wire sits at $h = 0.05\lambda$. By how many dB is its straight-up ($\theta = 0$) field below that of the same wire at $h = 0.25\lambda$?

   Δ = -10.2 dB

   $$h = 0.05\lambda: \quad 2\sin(2\pi(0.05)) = 2\sin(18^\circ) = 0.618$$
   $$h = 0.25\lambda: \quad 2\sin(2\pi(0.25)) = 2\sin(90^\circ) = 2.00$$
   $$20\log_{10}(0.618/2.00) = -10.2 \text{ dB}$$

   (b) Evaluate the vertical array factor at the horizon ($\theta = 90^\circ$) for an arbitrary height h. What does the result tell you?

   AF = 2

   $$2|\cos(kh\cos 90^\circ)| = 2|\cos(0)| = 2 \quad \text{for every } h$$

   A vertical element and its image always add in phase along the horizon, at any height, including zero. That is why a vertical antenna works while standing on the ground.

   (c) In one or two sentences, explain why a horizontal wire lying directly on a metal deck radiates essentially nothing.

   A horizontal (tangential) current images with the opposite sign, so the deck creates an equal and opposite current directly beneath the wire. As the height goes to zero the two are collocated and cancel everywhere, and the radiation resistance collapses with them, so nearly all of the input power is dissipated rather than radiated.

   (d) You must put up a field-expedient antenna on a large metal roof for horizon-to-horizon communication, and you can only get about 0.05λ of height. Which orientation do you choose, and why?

   Vertical. At 0.05λ a horizontal wire is fighting its own image and loses more than 10 dB of the little it radiates, and perfect ground puts a null on the horizon for horizontal polarization at *every* height, which is exactly where the traffic is. A vertical element gets the in-phase image for free, keeps its pattern maximum on the horizon, and needs no height at all.

3. **The full loss budget of a magnetic loop.** A single-turn circular loop of radius $a = 0.30$ m is made of copper wire of radius $b = 3$ mm and operated at $f = 14.2$ MHz. At that frequency the copper surface resistance is $R_s = 0.98$ mΩ per square. Recall $R_r = 20\pi^2\,(C/\lambda)^4$ and $R_{\text{ohmic}} = (C/2\pi b)\,R_s$ for a single turn.

   (a) Compute C/λ and confirm the loop is electrically small.

   $$\lambda = \frac{3\times 10^8}{14.2\times 10^6} = 21.1\text{ m}$$
   $$C = 2\pi a = 1.885\text{ m}$$
   $$C/\lambda = 0.0892$$

   C/λ = $\boxed{0.0892}$

   Below the 0.1λ rule of thumb, so the current is essentially uniform around the loop and it behaves as a magnetic dipole.

   (b) Find the radiation resistance.

   $$R_r = 20\pi^2(0.0892)^4 = 197.4\left(6.34\times 10^{-5}\right)$$
   $$= 0.0125\,\Omega = 12.5\text{ m}\Omega$$

   R_r = $\boxed{12.5\text{ m}\Omega}$

   (c) Find the ohmic resistance and the radiation efficiency. Express the efficiency in dB and give the gain in dBi.

   $$R_{\text{ohmic}} = \frac{C}{2\pi b}R_s = \frac{a}{b}R_s$$
   $$= 100\,(0.98\text{ m}\Omega) = 0.098\,\Omega$$
   $$\eta_{\text{rad}} = \frac{R_r}{R_r + R_{\text{ohmic}}} = \frac{0.0125}{0.1105} = 0.113$$
   $$10\log_{10}(0.113) = -9.5\text{ dB}$$
   $$G = 1.76 - 9.5 = -7.7\text{ dBi}$$

   η_{rad} = $\boxed{11.3\%}$

   G = $\boxed{-7.7\text{ dBi}}$

   The loop's own copper burns nearly eight times what it radiates.

   (d) The same loop is a respectable receiving antenna at 14.2 MHz but a poor transmitting one. Explain the difference in one or two sentences.

   On transmit, 89% of your power heats the wire, and the tiny radiation resistance forces enormous circulating current. On receive, what matters is signal-to-noise, and below VHF the external atmospheric noise picked up by the antenna dominates the receiver's own noise – attenuating signal and sky noise together by 9.5 dB leaves the ratio essentially unchanged.

4. **Loop or dipole?** Continue with the loop of the previous problem.

(a) A direction-finding team rotates a small loop to take a bearing. Why do they turn the loop for a *null* rather than for a peak, and where does that null point relative to the loop?

The small-loop pattern is $|F| = \sin\theta$ with the null along the loop *axis* – straight through the hole – and the maximum in the plane of the loop. A null is sharp: the response changes quickly with a degree of rotation, so the bearing is precise. The peak is broad and flat, so a peak-seeking bearing would be several degrees wide. When the null is deepest, the transmitter lies along the loop axis.

(b) A vertical half-wave dipole and a small horizontal loop are both mounted on a mast. What polarization does each radiate toward a distant receiver on the horizon?

The vertical dipole radiates E_θ – vertical polarization. The horizontal loop is a magnetic dipole pointing straight up; its field is E_ϕ, which lies parallel to the ground at the horizon, so it radiates horizontal polarization. Same donut pattern, orthogonal polarizations: cross them and the polarization loss factor from Lesson 3 takes the link apart.

(c) Rewind the loop with $N = 20$ turns. Radiation resistance scales as N^2 and ohmic resistance as N. Find the new efficiency.

η_{rad} = $\boxed{71.8\%}$

$$
\begin{aligned}
R_r &= 20^2\,(0.0125) = 5.00\,\Omega \\
R_{\text{ohmic}} &= 20\,(0.098) = 1.96\,\Omega \\
\eta_{\text{rad}} &= \frac{5.00}{6.96} = 0.718
\end{aligned}
$$

71.8%, a loss of only 1.4 dB : gain $= 1.76 - 1.4 = 0.4$ dBi .

(d) If turns are that effective, why is a 20-turn loop still not the antenna you would choose to transmit with? Name two reasons.

(1) The N^2 scaling assumes the turns do not interact. Packed turns raise the loss through proximity effect and distributed capacitance, so the real efficiency falls short of the ideal number. (2) Adding turns adds inductance and stored energy without making the antenna electrically bigger, so the Q climbs and the usable bandwidth shrinks – and the tuning capacitor sees kilovolts. A resonant loop with $C \approx 1\lambda$ solves all of it: about 100 to $130\,\Omega$ of real, radiating impedance and roughly 3.1 dBi, with its maximum along the axis instead of in the plane.

5. **The price of being small.** Stay with the single-turn loop of radius $a = 0.30$ m at 14.2 MHz ($\eta_{\text{rad}} = 0.113$). The Chu limit from Lesson 3 gives the radiation quality factor of any antenna that fits inside a sphere of radius a as $Q \gtrsim 1/(ka)^3 + 1/(ka)$, and the fractional bandwidth is roughly $1/Q$.

   (a) Compute ka, the Chu Q, and the resulting bandwidth in kHz.

$$k = \frac{2\pi}{21.1} = 0.297\ \text{rad/m}, \quad ka = 0.0892$$

$$Q \gtrsim \frac{1}{(0.0892)^3} + \frac{1}{0.0892} = 1409 + 11 = 1420$$

$$\text{FBW} \approx \frac{1}{Q} = 7.0 \times 10^{-4} = 0.070\%$$

$$\Delta f = 0.00070\,(14.2\ \text{MHz}) = 10\ \text{kHz}$$

Q = $\boxed{\approx 1420}$

Δf = $\boxed{10\ \text{kHz}}$

   (b) A single-sideband voice channel needs about 3 kHz. Does the loop pass it, and what does your answer imply about operating across a 350 kHz amateur band?

   Yes. 10 kHz comfortably covers one 3 kHz voice channel, so the loop is usable on any single frequency. Moving elsewhere in a 350 kHz band means retuning the loop capacitor, because the match degrades within a channel or two. The size reduction is paid for in retuning.

   (c) Measured on a VNA, this loop shows a $2{:}1$ VSWR bandwidth several times wider than your Chu estimate. Why? Is that good news?

   Loss lowers Q. The Chu number is the *radiation* Q; the loaded Q is reduced roughly in proportion to the efficiency, $Q_{\text{loaded}} \approx \eta_{\text{rad}} Q \approx 0.113(1420) \approx 160$, which widens the match to about 0.6%, or 90 kHz. It is not good news: the extra bandwidth is bought by dissipating power in copper. A $50\ \Omega$ resistor is matched at every frequency and radiates nothing at all.

   (d) State the size-bandwidth trade as a rule of thumb, in one sentence.

   Shrinking an antenna below about $\lambda/2\pi$ in radius drives the stored near-field energy far above the radiated energy, so Q rises roughly as $1/(ka)^3$ and the usable bandwidth falls with the cube of the size – while the radiation resistance falls too, so you lose efficiency and bandwidth together.

Documentation: ____